
Department of Mathematics and Statistics,
Boston University,
665 Commonwealth Avenue, Room 417,
Boston, MA 02215, USA.

Email: achatter@bu.edu
Website: anirbanc96.github.io/main/
GitHub: [anirbanc96](https://github.com/anirbanc96)

Employment

Assistant Professor , <i>Department of Mathematics and Statistics, Boston University.</i> <i>Boston, Massachusetts, United States.</i>	2026 - Present
William H. Kruskal Instructor , <i>Department of Statistics, The University of Chicago.</i> <i>Chicago, Illinois, United States.</i>	2025 - 2026

Education

Ph.D. in Statistics , <i>The Wharton School, University of Pennsylvania, Philadelphia, United States.</i> Advisor: Bhaswar B. Bhattacharya.	2025
Master of Statistics , <i>Indian Statistical Institute, Kolkata, India.</i> First Division with Distinction, Specialization: Probability.	2020
Bachelor of Statistics (Hons.) , <i>Indian Statistical Institute, Kolkata, India.</i> First Division with Distinction.	2018

Publications

-
- **Anirban Chatterjee**, Sayantan Choudhury, Rohan Hore. One-shot Conditional Sampling: MMD meets Nearest Neighbors. *To appear in Proceedings of The 43rd International Conference on Machine Learning (ICML), 2026.*
 - **Anirban Chatterjee**, Soham Dan, Bhaswar B. Bhattacharya. Higher-Order Graphon Theory: Fluctuations, Degeneracies, and Inference. *The Annals of Statistics, Volume 54, No. 4, 2136 - 2163, 2026.*
 - **Anirban Chatterjee**, Aaditya Ramdas. A Martingale Kernel Two-Sample Test. *In Proceedings of The 37th International Conference on Algorithmic Learning Theory (ALT), PMLR 313:1-44, 2026.* [Elegant Paper Badge.](#)
 - **Anirban Chatterjee**, Jiaoyang Huang. Fluctuation of the Largest Eigenvalue of a Kernel Matrix with application in Graphon-based Random Graphs. *The Annals of Applied Probability, Volume 35, No. 6, 4244-4281, 2025.*
 - **Anirban Chatterjee**, Bhaswar B. Bhattacharya. Boosting the Power of Kernel Two-Sample Tests. *Biometrika, Volume 112, Issue 1, 2025.*
 - **Anirban Chatterjee**, Sagnik Nandy, Ritwik Sadhu. Detecting Planted Partition in Sparse Multilayer Networks. *Information and Inference: A Journal of the IMA, Volume 13, Issue 3, 2024.*
 - Abhinav Chakraborty, **Anirban Chatterjee**, Abhinandan Dalal. PrIsing: Privacy-Preserving Peer Effect Estimation via Ising Model. *In Proceedings of The 27th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 238:2692-2700, 2024.*
 - Bhaswar B. Bhattacharya, **Anirban Chatterjee**, Svante Janson. Fluctuations of Subgraph Counts in Graphon based Random Graphs. *Combinatorics, Probability and Computing, Volume 32, Issue 3, 2023.*
 - **Anirban Chatterjee**, Rajat Subhra Hazra. Spectral Properties for the Laplacian of a Generalized Wigner matrix. *Random Matrices: Theory and Applications, Volume 11, No. 03, 2022.*

Preprints

- Rohan Hore, **Anirban Chatterjee**, Sayantan Choudhury. Multi-Source Conformal Prediction: Leveraging Heterogeneity via Localization. [arXiv:2609.14531](#), 2026.
- **Anirban Chatterjee**, Rina Foygel Barber. A Ranking Approach for Measuring Calibration. [arxiv:2609.13100](#), 2026.
- Sanchayan Bhowal, **Anirban Chatterjee**, Somabha Mukherjee. Scaling Limits for Ising Models on Inhomogeneous Random Graphs and Applications. [arXiv:2608.12804](#), 2026.
- **Anirban Chatterjee**, Ziang Niu, Bhaswar B. Bhattacharya. Conditional Mean Independence and Global Sensitivity Analysis using Nearest Neighbor Graphs. [arXiv:2607.04692](#), 2026.
- Somabha Mukherjee, Sanchayan Bhowal, **Anirban Chatterjee**, Bhaswar B. Bhattacharya. Ising Models on Inhomogeneous Random Graphs: Inference, Local Asymptotic Minimality, and Limit of Experiments. [arXiv:2606.07065](#), 2026.
- Sayak Chatterjee, **Anirban Chatterjee**, Abhinav Chakraborty, Bhaswar B. Bhattacharya. Asymptotic Normality of Subgraph Counts in Sparse Inhomogeneous Random Graphs. [arXiv:2512.12937](#), 2025. (Minor revision at *Bernoulli*.)
- **Anirban Chatterjee**, Ziang Niu, Bhaswar B. Bhattacharya. A Kernel-Based Conditional Two-Sample Test Using Nearest Neighbors (with Applications to Calibration, Regression Curves, and Simulation-Based Inference). [arXiv:2407.16550](#), 2024.

Teaching and Mentoring

Instructor, Boston University.

- CAS MA 781: Estimation Theory - Fall 2026.

Instructor, The University of Chicago.

- STAT 20000: Elementary Statistics - Spring 2026.
- STAT 37779: Kernel Methods for Statistics and Machine Learning - Winter 2026.

Teaching Assistant, The Wharton School, University of Pennsylvania.

- STAT 4320: Mathematical Statistics - Spring 2022, Fall 2023.
- STAT 9610: Statistical Methodology - Fall 2022.

Graduate Mentor, *Undergraduate Research in Probability and Statistics (URPS)*, University of Pennsylvania, 2022.

Awards & Honors

- **Best Student Paper Award in Probability and Theory of Statistics & Data Science**, International Indian Statistical Association, 2024.
- **J. Parker Bursk Memorial Prize for Excellence in Research**, University of Pennsylvania, 2024.
- **George James Term Fund for The Bernoulli-IMS 11th World Congress in Probability and Statistics**, University of Pennsylvania, 2024.
- **Nominated for the Lawrence D. Brown Student Paper Award**, University of Pennsylvania, 2023.
- **George James Term Fund for the Joint Statistical Meetings**, University of Pennsylvania, 2023.
- **Nominated for the Prasanta Chandra Mahalanobis Gold Medal**, Indian Statistical Institute, 2020.
- **Dean's List for Toppers**, Indian Statistical Institute, 2016 – 2020.

Industry Experience

Project Team Member, TCS ion, Performance Analytics for Large Scale Online Assessments, 2017.

Professional Service

- Reviewer for Journals: *The Annals of Statistics*, *Journal of the Royal Statistical Society: Series B*, *Journal of the American Statistical Association*, *Information and Inference: A Journal of the IMA*, *Bernoulli*, *Statistical Science*, *Stochastic Processes and their Applications*, *Journal of Business & Economic Statistics*, *Applied Probability Journals*, *SIAM Journal on Mathematics of Data Science*.

Technical Skills

- **Programming Languages:** R, Python.
- **Tools and Software:** LaTeX.

Conferences & Workshops

Recent Advances in Random Networks: Theory and Applications, IMSI, Chicago, USA, 2026.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

Workshop on Uncertainty Quantification and Reliability, IDEAL Special Program on High Dimensional and Complex Data Analysis, Evanston, USA, 2025.

- Topic: Detecting Miscalibration: A Nearest Neighbor Approach.

Annual Conference of International Indian Statistical Association, Cochin, India, 2024.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

Bernoulli-IMS 11th World Congress in Probability and Statistics, Bochum, Germany, 2024.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

ICSA-Canada Chapter 2024 Symposium Niagara Falls, Canada, 2024.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

Lawrence D. Brown Student Workshop, University of Pennsylvania, Philadelphia, USA, 2023.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

Joint Statistical Meeting, Toronto, Canada, 2023.

- Topic: Clustering Network Vertices in Sparse Contextual Multilayer Networks.

ASU Seminar, Indian Statistical Institute, Kolkata, India, 2023.

- Topic: Higher Order Graphon Theory: Fluctuations and Inference.

IMS Annual Meeting, London, UK, 2022.

- Topic: Fluctuations of Subgraph counts in Graphon based Random Graphs.

P.C.M. Gold Medal Presentation, Indian Statistical Institute, Kolkata, India, 2020.

- Topic: Graphon Estimation.